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أدنوك
ADNOC

ADNOC GROUP PROJECTS AND ENGINEERING

CIVIL CONSTRUCTION MATERIALS SAMPLING AND TESTING SPECIFICATION

Specification

APPROVED BY:

24/11/2021

NAME: Abdulmunim Al Kindy
TITLE: Executive Director PT&CS
EFFECTIVE DATE:

AGES-SP-01-009

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شركة بترول أبوظبي الوطنية
Abu Dhabi National Oil Company

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GROUP PROJECTS & ENGINEERING / PT&CS DIRECTORATE

CUSTODIAN	Group Projects & Engineering / PT&CS
ADNOC	Specification applicable to ADNOC & ADNOC Group Companies

REVISION HISTORY

DATE	REV. NO	PREPARED BY (Designation / Initial)	REVIEWED BY (Designation / Initial)	ENDORSED BY (Designation / Initial)	ENDORSED BY (Designation / Initial)
22-Sep-2021	1	Kuriakose Molelputhenpura / SE – Civil - AGP Digitally signed by Kuriakose Mathew DN: cn=Kuriakose Mathew, o=AGP, ou=ADNOC, email=kmathew@adnoc.ae, c=AE Date: 2021.10.05 10:35:27 +04'00'	Mahmoud Abdel Hakim / HOD Pipelines Eng. - GP&E Mahmoud Abdel Hakim Digitally signed by Mahmoud Abdel Hakim Date: 2021.10.11 12:13:36 +04'00' Reuben Yagambaram/ Mgr, Offshore Portfolio - GP&E Reuben Yagambaram Digitally signed by Reuben Yagambaram Date: 2021.10.12 11:50:53 +04'00'	Najem Qambar / VP Group Eng. - GPE 23/11/2021 Ali AlBreiki/ VP Upstream Projects – GP&E 23/11/2021	Ebraheem AlRomaithi / SVP – GP&E 23/11/2021

Group Projects & Engineering is the owner of this Specification and responsible for its custody, maintenance and periodic update.

In addition, Group Projects & Engineering is responsible for communication and distribution of any changes to this Specification and its version control.

This specification will be reviewed and updated in case of any changes affecting the activities described in this specification.

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INTER-RELATIONSHIPS AND STAKEHOLDERS

a. The following are inter-relationships for implementation of this Specification:
   i. ADNOC Upstream and ADNOC Downstream Industry, Marketing & Trading Directorate.
   ii. ADNOC Onshore, ADNOC Offshore, ADNOC Sour Gas, ADNOC Gas Processing. ADNOC LNG, ADNOC Refining, Fertil, Borouge, Al Dhafra Petroleum, Al Yasat
b. The following are stakeholders for the purpose of this Specification:
   i. ADNOC PT&CS Directorate
c. This Specification has been approved by the ADNOC PT&CS is to be implemented by each ADNOC Group company included above subject to and in accordance with their Delegation of Authority and other governance-related processes in order to ensure compliance.
d. Each ADNOC Group company must establish/nominate a Technical Authority responsible for compliance with this Specification.

DEFINITIONS

"**ADNOC**" means Abu Dhabi National Oil Company.

"**ADNOC Group**" means ADNOC together with each company in which ADNOC, directly or indirectly, controls fifty percent (50%) or more of the share capital.

"**Approving Authority**" means the decision-making body or employee with the required authority to approve Policies & Procedures or any changes to it.

"**Business Line Directorates**" or "**BLD**" means a directorate of ADNOC which is responsible for one or more Group Companies reporting to, or operating within the same line of business as, such directorate.

"**Business Support Directorates and Functions**" or "**Non- BLD**" means all the ADNOC functions and the remaining directorates, which are not ADNOC Business Line Directorates.

"**CEO**" means chief executive officer.

"**Group Company**" means any company within the ADNOC Group other than ADNOC.

"Specification" means this Civil Construction Materials Sampling and Testing specification.

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1 GENERAL

1.1 Introduction

This specification provides the minimum requirements for sampling and testing of Civil construction materials in ADNOC projects.

1.2 Purpose

This specification contains the minimum requirements for the accreditation procedures, type and frequency of tests, personnel requirements, quality control measures and equipment calibration in order for the CONTRACTOR or its SUBCONTRACTOR to establish a laboratory with the quantity and type of equipment required to perform the tests. The purpose of the laboratory testing is to verify that the earthwork, bituminous asphalt materials, paving, concrete, concrete materials, reinforcement and anchor bolts conform to the Project Drawings and Specifications.

Civil construction materials not covered under this specification shall be tested in accordance with other COMPANY / Project Specifications and material MANUFACTURER / SUPPLIER recommendations. All procured items should be covered by the manufacturer's accreditation.

1.3 Definitions and Abbreviations

1.3.1 Definitions

The following defined terms are used throughout this specification:

'**[PSR]**' indicates a mandatory Process Safety Requirement

"**COMPANY**" means ADNOC, ADNOC Group or an ADNOC Group Company, and includes any agent or consultant authorized to act for, and on behalf of the COMPANY.

"**CONTRACTOR**" means the parties that carry out all or part of the design, engineering, procurement, construction, commissioning or management for ADNOC projects. CONTRACTOR includes its approved MANUFACTURER(s), SUPPLIER(s), SUB-SUPPLIER(s) and SUB-CONTRACTOR(s).

"**MANUFACTURER**" means the Original Equipment Manufacturer (OEM) or MANUFACTURER of one or more of the component(s) which make up a sub-assembly or item of equipment assembled by the main SUPPLIER or his nominated SUB-SUPPLIER.

'may' means a permitted option

'shall' indicates mandatory requirements

'should' means a recommendation

"**SUB-CONTRACTOR**" means any party engaged by the CONTRACTOR to undertake any assigned work on their behalf. COMPANY maintains the right to review all proposed SUB-CONTRACTORs; this right does not relieve the CONTRACTOR of their obligations under the Contract, nor does it create any contractual relationship between COMPANY and the SUB-CONTRACTOR.

"SUPPLIER" means the party entering into a Contract with COMPANY to provide the materials, equipment, supporting technical documents and/or drawings, guarantees, warranties and/or agreed services in accordance with the requirements of the purchase order and relevant specification(s). The term SUPPLIER includes any legally appointed successors and/or nominated representatives of the SUPPLIER.

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"**SUB-SUPPLIER**" means the sub-contracted SUPPLIER of equipment sub-components software and/or support services relating to the equipment / package, or part thereof, to be provided by the SUPPLIER. COMPANY maintains the right to review all proposed SUB-SUPPLIERS, but this right does not relieve the SUPPLIER of their obligations under the Contract, nor does it create any contractual relationship between COMPANY and any individual SUB-SUPPLIER.

1.3.2 Abbreviations

The abbreviations used throughout this specification are shown in Table 1.1

Table 1.1 List of Abbreviations

Abbreviations	
AASTHO	American Association of State Highways and Transportation Official
ACI	American Concrete Institute
ASTM	American Society for Testing and Materials
BSI	British Standards Institution
DIN	Deutshes Institut Fir Normung E. V.
ISO	International Organization for Standardization
OSHA	Occupational Safety and Health Administration
RILEM	Reunion Internationale Des Laboratories D'essais Et De Recherches Sur Les Materiaux Et Les Constructions
UKAS	United Kingdom Accreditation Service
UL	Underwriters Laboratories, Inc.

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SECTION A – GENERAL

2 REFERENCE DOCUMENTS

2.1 International Codes and Standards

The following Codes and Standards shall form a part of this specification. When an edition date is not indicated for a Code or Standard, the latest edition in force at the time of the contract award shall apply.

AMERICAN ASSOCIATION OF STATE HIGHWAYS AND TRANSPORTATION OFFICIAL (AASHTO)

AASHTO T 27	Standard Method of Test for Sieve Analysis of Fine and Coarse Aggregates
AASHTO T 37	Standard Method of Test for Sieve Analysis of Mineral Filler for Hot Mix Asphalt (HMA)
AASHTO T 96	Standard Method of Test for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
AASHTO T 236	Standard Method of Test for Direct Shear Test of Soils under Consolidated Drained Conditions
AASHTO T 259	Standard Method of Test for Resistance of Concrete to Chloride Ion Penetration
AASHTO T 260	Standard Method of Test for Sampling and Testing for Total Chloride Ion in Concrete and Concrete Raw Materials
AASHTO T 277	Standard Method of Test for Electrical Indication of Concrete's Ability to Resist Chloride Ion Penetration
AASHTO R 66	Standard Practice for Sampling Asphalt Materials
AASHTO R 97	Standard Practice for Sampling Asphalt Mixtures

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 304R	Guide for Measuring, Mixing, Transporting and Placing Concrete

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products

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ASTM A615/A615M	Standard Specification for Deformed and Plain Carbon Steel Bars for Concrete Reinforcement
ASTM A751	Standard Test Methods, and Practices for Chemical Analysis of Steel Products
ASTM A880.	Standard Practice for Criteria for Use in Evaluation of Testing Laboratories and Organizations for Examination and Inspection of Steel, Stainless Steel, and Related Alloys
ASTM C29/C29M	Standard Test Method for Bulk Density (Unit Weight) and Voids in Aggregate.
ASTM C31/C31M	Standard Practice for Making and Curing Concrete Test Specimens in the Field
ASTM C33/C33M	Standard Specification for Concrete Aggregates
ASTM C40/C40M	Standard Test Method for Organic Impurities in Fine Aggregates for Concrete
ASTM C87/C87M	Standard Test Method for Effect of Organic Impurities in Fine Aggregate on Strength of Mortar
ASTM C88/C88M	Standard Test Method for Soundness of Aggregates by use of Sodium Sulphate or Magnesium Sulphate
ASTM C94/C94M	Standard Specification for Ready-Mixed Concrete.
ASTM C109/C109M	Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (Using 2 Inch or 50 mm Cube Specimens)
ASTM C117	Standard Test Method for Materials Finer than 75 Micrometer (No. 200) Sieve in Mineral Aggregates by Washing
ASTM C123	Standard Test Method for Lightweight Particles in Aggregate
ASTM C127	Standard Test Method for Relative Density (Specific Gravity) and Absorption of Coarse Aggregate.
ASTM C128	Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Fine Aggregate.
ASTM C131/C131M	Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
ASTM C136	Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates

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ASTM C142/C142M	Standard Test Method for Clay Lumps and Friable Particles in Aggregates
ASTMC143/C143M	Standard Test Method for Slump of Hydraulic Cement Concrete
ASTM C150/C150M	Standard Specification for Portland Cement
ASTMC172/C172M	Standard Practice for Sampling Freshly Mixed Concrete
ASTM C183	Standard Practice for Sampling and the Amount of Testing of Hydraulic Cement
ASTM C186	Standard Test Method for Heat of Hydration of Hydraulic Cement
ASTM C188	Standard Test Method for Density of Hydraulic Cement
ASTM C191	Standard Test Methods for Time of Setting of Hydraulic Cement by Vicat Needle
ASTM C192/C192M	Standard Practice for Making and Curing Concrete Test Specimens in the Laboratory
ASTM C289	Standard Test Method for Potential Alkali-Silica Reactivity of Aggregates (Chemical Method)
ASTM C295/C295M	Standard Guide for Petrographic Examination of Aggregates for Concrete
ASTM C457/C457M	Standard Test Method for Microscopical Determination of Parameters of the Air-Void System in Hardened Concrete
ASTM C494/C494M	Standard Specification for Chemical Admixtures for Concrete
ASTM C535	Standard Test Method for Resistance to Degradation of Large-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
ASTM C 579	Standard Test Methods for Compressive Strength of Chemical-Resistant Mortars, Grouts, Monolithic Surfacings, and Polymer Concretes
ASTM C642	Standard Test Method for Density, Absorption, and Voids in Hardened Concrete
ASTM C823/C823M	Standard Practice for Examination and Sampling of Hardened Concrete in Constructions
ASTM C856	Standard Practice for Petrographic Examination of Hardened Concrete
ASTM C989/C989M	Standard Specification for Slag Cement for Use in Concrete and Mortars
ASTM C1064/C1064M	Standard Test Method for Temperature of Freshly Mixed Hydraulic-Cement Concrete

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ASTM C1077	Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation
ASTM C1084	Standard Test Method for Portland-Cement Content of Hardened Hydraulic-Cement Concrete
ASTM C1105/C1105M	Standard Test Method for Length Change of Concrete Due to Alkali-Carbonate Rock Reaction
ASTM C1152/C1152M	Standard Test Method for Acid-Soluble Chloride in Mortar and Concrete
ASTM C1202	Standard Test Method for Electrical Indication of Concrete's Ability to Resist Chloride Ion Penetration
ASTM C1218/C1218M	Standard Test Method for Water-Soluble Chloride in Mortar and Concrete
ASTM C1260	Standard Test Method for Potential Alkali Reactivity of Aggregates (Mortar-Bar Method)
ASTM C1582/C1582M	Standard Specification for Admixtures to Inhibit Chloride-Induced Corrosion of Reinforcing Steel in Concrete
ASTM D5/D5M	Standard Test Method for Penetration of Bituminous Materials
ASTM D75/D75M	Standard Practice for Sampling Aggregates
ASTM D422	Standard Test Method for Particle-Size Analysis of Soils
ASTM D512	Standard Test Method for Chloride Ion in Water
ASTM D516	Standard Test Method for Sulfate Ion in Water
ASTM D546	Standard Test Method for Sieve Analysis of Mineral Filler for Asphalt Paving Mixtures
ASTM D854	Standard Test Methods for Specific Gravity of Soil Solids by Water Pycnometer
ASTM D979/D979M	Standard Practice for Sampling Bituminous Paving Mixtures
ASTM D1188	Standard Test Method for Bulk Specific Gravity and Density of Compacted Bituminous Mixtures Using Coated Samples
ASTM D1293	Standard Test Methods for pH of Water
ASTM D1411	Standard Test Methods for Water-Soluble Chlorides Present as Admixtures in Graded Aggregate Road Mixes

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ASTM D1556/D1556M	Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method
ASTM D1557	Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft3 (2,700 kN-m/m3)).
ASTM D1856	Standard Test Method for Recovery of Asphalt from Solution by Abson Method
ASTM D2041/D2041M	Standard Test Method for Theoretical Maximum Specific Gravity and Density of Asphalt Mixtures
ASTM D2070	Standard Test Method for Thermal Stability of Hydraulic Oils
ASTM D2167	Standard Test Method for Density and Unit Weight of Soil In-Place by the Rubber Balloon Method
ASTM D2170/D2170M	Standard Test Method for Kinematic Viscosity of Asphalts
ASTM D2172/D2172M	Standard Test Methods for Quantitative Extraction of Asphalt Binder from Asphalt Mixtures
ASTM D2216	Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass
ASTM D2487	Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)
ASTM D2488	Standard Practice for Description and Identification of Soils (Visual-Manual Procedures)
ASTM D2726/D2726M	Standard Test Method for Bulk Specific Gravity and Density of Non-Absorptive Compacted Asphalt Mixtures
ASTM D3080	Standard Test Method for Direct Shear Test of Soils under Consolidated Drained Conditions
ASTM D3203/D3203M	Standard Test Method for Percent Air Voids in Compacted Asphalt Mixtures
ASTM D3665	Standard Practice for Random Sampling of Construction Materials
ASTM D3666	Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials
ASTM D3740	Standard Practice for Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction

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ASTM D3856	Standard Guide for Management Systems in Laboratories Engaged in Analysis of Water
ASTM D4318	Standard Test Method for Liquid Limit, Plastic Limit, and Plasticity Index of Soils
ASTM D4541	Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers
ASTM D4767	Standard Test Method for Consolidated Undrained Triaxial Compression Test for Cohesive Soils
ASTM D5404/D5404M	Standard Practice for Recovery of Asphalt from Solution Using the Rotary Evaporator
ASTM D6913/D6913M	Standard Test Methods for Particle-Size Distribution (Gradation) of Soils Using Sieve Analysis.
ASTM D6938	Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth).
ASTM D7906	Standard Practice for Recovery of Asphalt from Solution Using Toluene and the Rotary Evaporator
ASTM E329	Standard Specification for Agencies Engaged in Construction Inspection, Testing or Special Inspection
ASTM F606/F606M	Standard Test Methods for Determining the Mechanical Properties of Externally and Internally Threaded Fasteners, Washers, Direct Tension Indicators, and Rivets

BRITISH STANDARDS INSTITUTION (BSI)

BS 598-1	Sampling and Examination of Bituminous Mixtures for Roads and Other Paved Areas Part 1: Methods for the Measurement of the Rate of Spread of Coated Chippings and the Temperature of Bituminous Mixtures Using Non-Contact Temperature-Measuring Devices and for the Assessment of the Compaction Performance of a Roller
BS 812-2	Testing of Aggregates
BS 812-103.2	Testing aggregates - Part 103: Method for determination of particle size distribution - Section 103.2 Sedimentation test
BS 812-121	Testing aggregates - Part 121: Method for determination of soundness
BS 1377-1	Methods of Test for Soils for Civil Engineering Purposes - Part 1: General requirements and Sample preparation

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BS 1377-3	Methods of test for Soils for Civil Engineering Purposes - Part 3: Chemical and Electrochemical Tests
BS 1377-4	Methods of Test for Soils for Civil Engineering Purposes - Part 4: Compaction-related Tests
BS 1377-9	Methods of Test for Soils for Civil Engineering Purposes - Part 9: In-Situ Tests
BS 1427	Guide to On-Site Test Methods for the Analysis of Waters
BS 1881- 122	Testing Concrete - Part 122: Method for Determination of Water Absorption
BS 1881- 124	Testing Concrete - Part 124: Methods of Analysis of Hardened Concrete
BS 1881- 128	Testing concrete Part 128. Methods for analysis of fresh concrete
BS 4449	Steel for the reinforcement of concrete. Weldable reinforcing steel. Bar, coil and decoiled product - Specification
BS 4482	Steel wire for the reinforcement of concrete products - Specification
BS 4483	Steel Fabric for the Reinforcement of Concrete – Specification
BS 4550-3.8	Methods of Testing Cement - Part 3: Physical Tests - Section 3.8 Test for Heat of Hydration
BS 6200-3.0	Sampling and Analysis of Iron, Steel and Other Ferrous Metals - Part 3.0: Reference Methods of Analysis – Summary of Methods
BS 8500-1	Concrete - Complementary British Standard to BS EN 206 Part 1: Method of specifying and guidance for the specifier
BS 8500-2	Concrete - Complementary British Standard to BS EN 206 Part 2: Specification for constituent materials and concrete
BS 594987	Asphalt for Roads and other Paved Areas - Specification for Transport, Laying, Compaction and Product Type Testing Protocols
BS DD 216	Method for Determination of Chloride Content of Fresh Concrete

EURO CODES (BS EN)

BS EN 196-1	Methods of testing cement - Part 1: Determination of Strength
BS EN 196-2	Method of Testing Cement - Part 2: Chemical analysis of cement
BS EN 196-7	Methods of Testing Cement Part 7: Methods of Taking and Preparing Samples of Cement

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BS EN 197-1	Cement Part 1: Composition, specifications and conformity criteria for common cements
BS EN 206	Concrete - Specification, performance, production and conformity
BS EN 450-1	Fly Ash for Concrete - Part 1: Definition, Specifications and conformity criteria
BS EN 480	Admixtures for concrete, mortar and grout. Test methods
BS EN 932-1	Tests for General Properties of Aggregates Part 1: Methods for Sampling
BS EN 932-6	Tests for General Properties of Aggregates - Part 6: Definitions of Repeatability and Reproducibility
BS EN 933-1	Tests for geometrical properties of aggregates - Part 1: Determination of particle size distribution - Sieving method
BS EN 933-3	Tests for geometrical properties of aggregates - Part 3: Determination of particle shape - Flakiness index
BS EN 933 - 7	Tests for Geometrical Properties of Aggregates - Part 7: Determination of Shell Content - Percentage of Shells in Coarse Aggregates
BS EN 1008	Mixing water for concrete - Specification for sampling, testing and assessing the suitability of water, including water recovered from processes in the concrete industry, as mixing water for concrete
BS EN 1097-2	Tests for Mechanical and physical Properties of Aggregates - Part 2: Methods for the Determination of resistance to Fragmentation
BS EN 1097-3	Tests for Mechanical and physical Properties of Aggregates - Part 3: Determination of Loose Bulk Density and Voids
BS EN 1097-6	Tests for mechanical and physical properties of aggregates - Part 6: Determination of particle density and water absorption
BS EN 1367-2	Tests for thermal and weathering properties of aggregates - Part 2: Magnesium sulphate test
BS EN 1744-1	Tests for chemical properties of aggregates - Part 1: Chemical analysis.
BS EN 12350-1	Testing fresh Concrete - Part 1: Sampling and common apparatus
BS EN 12350-2	Testing fresh Concrete - Part 2: Slump test
BS EN 12390-1	Testing hardened concrete - Part 1: Shape, dimensions and other requirements for specimens and moulds

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BS EN 12390-2	Testing hardened concrete - Part 2: Making and curing specimens for strength tests
BS EN 12390-3	Testing hardened concrete - Part 3: Compressive strength of test specimens
BS EN 12390-7	Testing hardened concrete - Part 7: Density of hardened concrete
BS EN 12390 - 8	Testing Hardened Concrete - Part 8: Depth of Penetration of Water Under Pressure
BS EN 12504-1	Testing concrete in structures Part 1: Cored specimens - Taking, examining and testing in compression
BS EN 12620	Aggregates for concrete
BS EN 12697-1	Bituminous Mixtures - Test Methods Part 1: Soluble Binder Content
BS EN 12697-2	Bituminous Mixtures - Test Methods Part 2: Determination of bulk density of bituminous specimens
BS EN 12697-6	Bituminous Mixtures - Test Methods Part 6: Determination of Bulk Density of Bituminous Specimens
BS EN 12697-10	Bituminous Mixtures - Test Methods Part 10: Compactibility
BS EN 12697-11	Bituminous Mixtures - Test Methods Part 11: Determination of the Affinity between Aggregate and Bitumen
BS EN 12697-14	Bituminous Mixtures - Test Methods Part 14: Water Content
BS EN 12697-27	Bituminous Mixtures - Test Methods Part 27: Sampling
BS EN 12697-28	Bituminous Mixtures - Test Methods Part 28: Preparation of Samples for Determining Binder Content, Water Content and Grading
BS EN 12697-34	Bituminous Mixtures - Test Methods Part 34: Marshall Test
BS EN 13055	Lightweight aggregates - Lightweight aggregates for concrete, mortar and grout
BS EN 15167-1	Ground granulated blast furnace slag for use in concrete, mortar and grout - Part 1: Definitions, specifications and conformity criteria
BS EN 15167-2	Ground granulated blast furnace slag for use in concrete, mortar and grout - Part 2: Conformity evaluation
BS EN ISO/IEC 17025	General requirements for the competence of testing and calibration laboratories

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BS EN ISO 6892-1	Metallic Materials - Tensile Testing - Part 1: Method of Test at Room Temperature

DEUTSHES INSTITUT FIR NORMUNG E. V. (DIN)

DIN 1048	Water Impermeability/Testing of Fresh Concrete

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 9000	Quality Management Systems – Fundamentals and vocabulary
ISO 9001	Quality Management Systems - Requirements
ISO 9004	Quality management - Quality of an organization - Guidance to achieve sustained success.
ISO/IEC Guide 2	Standardization and Related Activities - General Vocabulary
ISO/IEC 17025	General Requirements for the Competence of Testing and Calibration Laboratories
ISO Guide 30	Reference Materials - Selected terms and definitions
ISO 19011	Guidelines for auditing management systems

REUNION INTERNATIONALE DES LABORATORIES D'ESSAIS ET DE RECHERCHES SUR LES MATERIAUX ET LES CONSTRUCTIONS (RILEM)

RILEM CPC 11.3	Absorption of Water by concrete by Immersion Under Vacuum

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

OSHA Parts 1910 and 1926	Code of Federal Regulation

UNDERWRITERS LABORATORIES, INC. (UL)

UL 1709	Standard for Safety Rapid Rise Fire Tests of Protective Materials for Structural Steel

CONCRETE PLANT MANUFACTURER BUREAU (CPMB):

CPMB 100-00	Concrete Plant Standards

2.2 ADNOC Specifications

The following ADNOC general specifications, as available with respective Group Company and as applicable to the Project, shall form a part of this specification. When an edition date is not indicated for a document, the latest edition in force at the time of contract award shall apply.

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Concrete Supply and Construction Specification	AGES-SP-01-001
Anchor Bolts and Grouting Specification	AGES-SP-01-004
Roads and Paving – Design and Construction Specification	AGES-SP-01-005
Site Preparation, Earthworks and Bund Construction Specification	AGES-SP-01-006
Onshore Geotechnical Investigation	AGES-SP-01-010
Concrete Repair Specification	TBC

2.3 Standard Drawings

Not applicable.

2.4 Other References

Not applicable.

3 DOCUMENT PRECEDENCE

The specifications and codes referred to in this specification shall, unless stated otherwise, be the latest approved issue at the time of contract award.

It shall be the CONTRACTOR's responsibility to be, or to become, knowledgeable of the requirements of the referenced Codes and Standards.

The CONTRACTOR shall notify the COMPANY of any apparent conflict between this specification, the related data sheets, the Codes and Standards and any other specifications noted herein.

Resolution and/or interpretation precedence shall be obtained from the COMPANY in writing before proceeding with the design/manufacture.

In case of conflict, the order of document precedence shall be:

a. UAE Statutory requirements

b. ADNOC HSE Standards

c. Equipment datasheets and drawings

d. Project Specifications and standard drawings

e. Company Specifications

f. National / International Standards

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4 SPECIFICATION DEVIATION / CONCESSION CONTROL

Any technical deviations to the Purchase Order and its attachments including, but not limited to, the COMPANY's General Specifications shall be sought by the CONTRACTOR only through technical deviation request format. Technical deviation requests require COMPANY'S review/approval, prior to the proposed technical changes being implemented. Technical changes implemented prior to COMPANY approval are subject to rejection.

5 DESIGN CONSIDERATIONS / MINIMUM DESIGN REQUIREMENTS

Refer to Section B of this specification.

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SECTION B – TECHNICAL REQUIREMENTS

6 GENERAL REQUIREMENTS

6.1 Material and Equipment

The CONTRACTOR or its SUBCONTRACTOR(s) shall supply all material and equipment required to perform the tests as indicated in this specification.

6.2 Accreditation and Evaluation

a. The Testing Laboratory, including the facilities, personnel and equipment shall be accredited and qualified in accordance with ASTM E329 and other applicable ASTMs. As an alternative to these ASTMs, the Testing Laboratory shall be accredited by UKAS.

b. The Testing Laboratory shall operate a Quality System in accordance with ISO 9000, 9001, 9004 ISO 19011, ISO Guide 30, ISO/IEC Guide 2, and ISO/IEC 17025.

c. For the testing of concrete and concrete related materials the Testing Laboratory, procedures, personnel and equipment shall be in accordance with ASTM C1077.

d. For the testing of soil and rock the Testing Laboratory, procedures, personnel and equipment shall be in accordance with ASTM D3740.

e. For the testing of bituminous paving materials the Testing Laboratory, procedures, personnel and equipment shall be in accordance with ASTM D3666.

f. For the testing of reinforcing bars the Testing Laboratory, procedures, personnel and equipment shall be in accordance with ASTM A880.

g. For the testing of water the Testing Laboratory, procedures, personnel and equipment shall be in accordance with ASTM D3856.

h. For testing of fireproofing material, the Testing Laboratory, procedures, personnel, and equipment shall be in accordance with UL 1709.

i. For testing of bolts, nuts and washers, the testing laboratory procedures, personnel, and equipment shall be in accordance with ASTM F606 / F606M.

6.3 Sample Collection

The CONTRACTOR or its SUBCONTRACTOR(s) shall be responsible for acquiring, transporting and storing all samples required for testing.

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6.4 Coordination of Tests

It shall be the CONTRACTOR'S responsibility to coordinate all activities associated with the testing of materials for the testing required by this specification.

6.5 Manufacturer's Test Certificates

The validity period of Manufacturer's test certificate shall be maximum 6 months unless otherwise specified by COMPANY in the Contract documents.

6.6 Materials Not Covered Under This Specification

Civil construction materials not covered under this specification shall be tested in accordance with other COMPANY / Project Specifications and material MANUFACTURER / SUPPLIER test Certificates. All procured items should be covered by the manufacturer's accreditation.

7 CONCRETE TESTING AND FREQUENCY

The CONTRACTOR shall be responsible for ensuring that the following concrete and concrete related tests are performed at the frequency shown below and in accordance with the requirements of AGES-SP-01-001 and the Standards listed in Section 2.0:

MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
WATER AND ICE (Supplied from public water supply or a reverse osmosis plant)		
Chlorides	BS1427 Section 4.3 or Quantab test strips for test on truck deliveries	Weekly
Sulphates	BS 1427 Section 4.6 or Quantab Test Strips for tests on truck deliveries	Weekly
PH	BS 1427 Section 5.3	Monthly or every 1000m^3 whichever is more frequent.
Total Solids	BS 1427 Section 5.4	Monthly or every 1000m^3 whichever is more frequent.
Alkali Carbonates	BS 1427 Section 4.1	Monthly or every 1000m^3 whichever is more frequent.
Bicarbonates	BS 1427 Section 4.1	Monthly or every 1000m^3 whichever is more frequent.
CEMENT		
Standard physical and chemical properties for Ground Granulated Blast Furnace Slag Cement	BS EN 15167-1, BS EN 15167-2, BS EN 196-1, BS EN 196-2, BS EN 197-1, BS EN 197-2	Receipt and review of Manufacturer's Test Certificate for each delivery of cement to the site or to the batch plant.

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MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
Standard physical and chemical properties for Portland and High Slag blast furnace cements	BS EN 196-1, BS EN 196-2, BS EN 197-1, BS EN 197-2	Receipt and review of Manufacturer's Test Certificate for each delivery of cement to the site or to the batch plant.
Sampling	BS EN 196-7	As required by individual test.

MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY	
		COARSE AGGREGATE	FINE AGGREGATE
AGGREGATES			
Aggregates		2 sets of tests for each Single Size Coarse Aggregate	2 sets of tests for each Single Size fine Aggregate
Sampling	BS 812-101, BS 812-102, BS EN 932-1, BS EN 932-6 or ASTM D75	As required by individual test	As required by individual test.
Clay, silt and dust	BS EN 933-1	Every 2 weeks or 2,000 m^3 of concrete, whichever is more frequent	Every 1 week or 1,000 m^3 of concrete, whichever is more frequent.
Clay Lumps and friable particles	ASTM C142 ASTM C33	Every 2 weeks or 2,000 m^3 of concrete, whichever is more frequent	Every 1 week or 1,000 m^3 of concrete, whichever is more frequent.
Organic Impurities	ASTM C40 ASTM C87 BS EN 1744-1 Clause 15	(Test not applicable to coarse aggregate)	Every 4 weeks or 5,000 m^3 of concrete, whichever is more frequent
Chlorides as Cl	BS EN 1744-1 Clause 7, AASHTO T 259	Every 1 week or 1,000 m^3 of concrete, whichever is more frequent	Every 1 week or 1,000 m^3 of concrete, whichever is more frequent.
Sulphates as SO3	BS EN 1744-1 Clause 12, ASTM C1012	Every 1 week or 1,000 m^3 of concrete, whichever is more frequent.	Each 1 week or 1,000 m^3 of concrete, whichever is more frequent
Alkali Reactivity	ASTM C1260	At the start of the project and whenever there is a change in the source of supply	

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MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY	
		COARSE AGGREGATE	FINE AGGREGATE
Alkali-Silica Reactivity	BS 1881 Part 124 Method 10.1 ASTM C289	At the start of the project and whenever there is a change in the source of supply as necessary depending on the results of ASTM C1260 above	
Alkali-Carbonate Reactivity	ASTM C295, C1105 or BS 933-1	At the start of the project and whenever there is a change in the source of supply as necessary depending on the results of ASTM C1260 above	
Grading	BS EN 933-1 or ASTM C136 ASTM C33 BS EN 933-2	Every 4 weeks or 5,000 m^3 of concrete, whichever is more frequent.	Every 2 weeks or 2,000 m^3 of concrete, whichever is more frequent.
Relative density (S.S.D)	BS EN 1097-3	Every 4 weeks or 5,000 m^3 of concrete, whichever is more frequent.	Every 4 weeks or 5,000 m^3 of concrete, whichever is more frequent.
Water absorption	BS EN 1097-6 ASTM C127 ASTM C128	Every 2 weeks or 2,000 m^3 of concrete, whichever is more frequent.	Every 2 weeks or 2,000 m^3 of concrete, whichever is more frequent
Soundness (coarse aggregate)	ASTM C88 (19 to 9.5) mm and (9.5 to 4.75) mm (5 cycles: Sodium Sulphate) or BS 812-121, BS EN 1367-2	Every 8 weeks or 10,000 m^3 of concrete, whichever is more frequent.	Not applicable
Soundness (fine aggregate)	ASTM C88 (9.5 to 0.3) mm (5 cycles: Sodium Sulphate) or BS 812-121, BS EN 1367-2	Not applicable	Every 8 weeks or 10,000 m^3 of concrete, whichever is more frequent.
Flakiness index	BS EN 933-3	Every 4 weeks or 5,000 m^3 of concrete, whichever is more frequent.	(Test not applicable to fine aggregate)

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MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY	
		COARSE AGGREGATE	FINE AGGREGATE
Elongation index	ASTM C490	Every 4 weeks or 5,000 m^3 of concrete, whichever is more frequent.	(Test not applicable to fine aggregate)
10% Fines value	BS 812, Part 111 (Standard test size), BS EN 1097-2	Every 8 weeks or 10,000 m^3 of concrete, whichever is more frequent.	(Test not applicable to fine aggregate)
Los Angeles Abrasion	ASTM C131 (Standard Gradings B and C) ASTM C 535	Every 8 weeks or 10,000 m^3 of concrete, whichever is more frequent	(Test not applicable to fine aggregate)

MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
ADMIXTURES		
Standard physical and chemical properties of admixtures	ASTM C494 BS EN 480	Receipt and review of Manufacturer's Test Certificate for each delivery of admixture to the site
Corrosion Inhibitor	ASTM C1582/C1582M	Receipt and review of Manufacturer's Test Certificate for each delivery of admixture to the site
FRESH CONCRETE		
Sampling	BS EN 12350-1 ASTM C172	As required by individual test.
Cube making, including curing	BS EN 12390-2	As required for hardened concrete strength tests.
Slump	BS EN 12350-2 ASTM C143/C143M	One (1) test per truck load with additional test if any truck (whether tested or not) is unduly delayed.
Temperature	BS 8500-1, BS 8500-2, BS EN 206-1	One (1) test per truck load with additional test if any truck (whether tested or not) is unduly delayed.

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MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
Cement content	BS 1881-128	Every 300 m³ of concrete. Note: COMPANY may request test, to check calibration of batch plant controls
Water content	BS 1881-128	Every 300 m³ of concrete. Note: COMPANY may request test, to check calibration of batch plant controls
Chlorides	BS DD 216 Method B.3	As for Chlorides in Hardened Concrete. Sample to be taken from the same batch as will be used for the test on the hardened concrete.
HARDENED CONCRETE (Requirements for blinding concrete will be limited to Sampling and Strength(cubes))		
Sampling	BS EN12350-1 BS EN 12504-1	As per individual test.
Strength (cubes)	Manufactured and cured to BS EN 12390 Parts 1 and 2, tested to BS EN 12390 Part 3, Density determined to BS EN 12390 Part 7.	3 cubes for every 60 m³ Min. or 1 set per day that concrete is placed, one tested at 7 days and two tested at 28 days.
Strength (drilled cores)	BS EN 12504-1 BS 1881: Part 124 or ASTM C42/C42M and ACI 214.4R-10	As required if cube tests do not meet acceptance criteria
Water Penetration (cubes)	DIN 1048 150mm cubes Cured a) 28 days with water, b) 14 days water then 14 days air. BS EN 12390-8	One test (3 specimens) every month for the first three months of production, for each grade of reinforced concrete, then one test (3 specimens) in every three months for each item a) and item b). One test result shall be the mean test result of three specimens.
Porosity	RILEM CPC 11.3 150mm dia x 75mm long cores from a) trial panel at 28 days and b) cubes which are water cured for 28 days	One test (3 specimens) every month for the first three months of production, for each grade of reinforced concrete, then one test (3 specimens) in every three months for each item a) and item b). One test result shall be the mean test result of three specimens.

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MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
Chloride Permeability	AASHTO T 259 100mm dia x 50mm long cores from a) trial panel at28 days and b) cubes which are water cured for 28 days	One test (3 specimens) every month for the first three month of production, then in every three months one test (3 specimens) for each item a) and item b).
Water Absorption	BS 1881 Part 122 or ASTM C642 75mm dia x 75mm long cores from a) trial panel at 28 days and b) cubes which are water cured for 28 days	One test (3 specimens) every month for the first three months of production, for each grade of reinforced concrete, then one test (3 specimens) in every three months for each item a) and item b). One test result shall be the mean test result of three specimens.
Total Chlorides	BS 1881 Part 124 Method 10.2	One test (3 specimens) every month for the first three months of production, for each grade of reinforced concrete, then one test (3 specimens) in every three months. One test result shall be the mean test result of three specimens.
Total Sulphates	BS 1881 Part 124 Method 10.3	One test (3 specimens) every month for the first three months of production, for each grade of reinforced concrete, then one test (3 specimens) in every three months. One test result shall be the mean test result of three specimens.
Air Voids	ASTM C457	One test (3 specimens) every month for the first three month of production, then in every three months one test (3 specimens). Note: This test is applicable to Air Entrained Concrete only.
Rapid chloride permeability test (RCPT)	ASTM C1202 (cores taken from cubes)	One test (4 specimens) every month for the first three months of production, for each grade of reinforced concrete, then one test (4 specimens) in every three months. One test result shall be the mean test result of four specimens.

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All samples shall be clearly marked with their identification and accurate records shall be submitted to COMPANY including the following data:

- Identification of test cube
- Date and time of pour and sampling
- Mix designation
- Location of sampled batched after placing
- Ambient temperature and humidity
- Cement type manufacture and quantity
- Volume of pour
- Concrete temperature (in position)
- Aggregate type source and proportions
- Admixture details
- Water/cement ratio
- Method of compaction
- Date of testing
- Results of test
- Weight of sample in grams
- Density in kg/m3
- Crushing load in Newton
- Crushing strength N/mm2
- Signature of person preparing cube
- Signature of person crushing cube

Note 1: The tests to be conducted on trial panel and those on the actual job mix shall be separately provided by the CONTRACTOR.

Note 2: Fibre reinforced concrete shall have similar tests to normal concrete.

8 REINFORCEMENT TESTING AND FREQUENCY

The CONTRACTOR shall be responsible for ensuring that the following reinforcement tests are performed at the frequency shown below and in accordance with the requirements of AGES-SP-01-001 and the Standards listed in Section 2.0:

MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
Standard physical and chemical Properties of reinforcing bars and steel fabric	BS 4449, BS 4482 & BS 4483	Receipt and review of Manufacturer's Test Certificate for each delivery of reinforcement bar or fabric to the site.

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MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
Mechanical Testing	Methods as specified in the relevant BS Standard	One test for a) Tensile Strength/Elongation b) Bend/Rebend On each 500 tons of each dia of reinforcement bar
Chemical Testing	Methods as specified in the relevant BS Standard	One test for a) Carbon, Phosphorous and sulphur per 500 tons of each dia of reinforcement bar b) Carbon equivalent per 100 tons of each dia of reinforcement bar for all reinforcement.

9 GROUT TESTING AND FREQUENCY

The CONTRACTOR shall be responsible for ensuring that the following grout tests are performed at the frequency shown below and in accordance with the requirements of AGES-SP-01-001, AGES-SP-01-004 and the Standards listed in Section 2.0:

MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
CEMENT BASED NON-SHRINK GROUT		
Compressive Strength	ASTM C109	Six cubes per day or for each 1m^3 or for each mechanical installation if critical, whichever is more stringent.
Water penetration	DIN 1048 (<5mm)	Once per week during grout installation.
Chloride permeability	AASHTO T 277 (<800 coulombs)	Once per week during grout installation.
EPOXY NON-SHRINK GROUT		
Compressive Strength	ASTM C579 Method B	Six cubes per day or for each 1m^3 or for each mechanical installation if critical, whichever is more stringent.
Other tests for epoxy grout shall be as per Manufacturer's requirement.		

10 CONCRETE COATING TESTING AND FREQUENCY

The CONTRACTOR shall be responsible for ensuring that the following coating tests are performed at the frequency shown below and in accordance with the requirements of AGES-SP-01-001 and the Standards listed in Section 2.0:

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MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
Adhesion	ASTM D4541	One pull out test which shall be the mean of three individual pull out tests for every 50 m^2 of coating layer. Individual pull out test shall be 0.5 m^2 area.
Other tests for coating shall be as per Manufacturer's requirement.		

11 SOIL TESTING AND FREQUENCY

The CONTRACTOR shall be responsible for ensuring that the following soil tests are performed at the frequency shown below and in accordance with the requirements of AGES-SP-01-006, AGES-SP-01-005 and the Standards listed in Section 2.0:

MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
Sampling	BS EN 932-1	As required by individual test.
Soil classification	ASTM D2487 and D2488	As required by AGES-SP-01-006 Site Preparation, Earthworks and Bund Construction Specification Any change in material source
Natural water content	ASTM D2216	As required by AGES-SP-01-006 Site Preparation, Earthworks and Bund Construction Specification Any change in material source, and one test for every 100 field density tests.
Density/Moisture (Modified Proctor Compaction test)	ASTM D1557	As required by AGES-SP-01-006 Site Preparation, Earthworks and Bund Construction Specification Any change in material source or description and monthly during earthworks
Particle size distribution, including hydrometer analysis where fraction passing 75-micron sieve exceeds 20%	ASTM D6913	As required by AGES-SP-01-006 Site Preparation, Earthworks and Bund Construction Specification Any change in material source or description and monthly during earthworks
Liquid limit, Plastic limit and Plasticity index	ASTM D4318	As required by AGES-SP-01-006 Site Preparation, Earthworks and Bund Construction Specification Any change in material source or description and monthly during earthworks

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MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
Angle of internal Friction	ASTM D4767 ASTM D3080	As required by AGES-SP-01-006 Site Preparation, Earthworks and Bund Construction Specification Any change in material source or description and monthly during earthworks
Chloride ion content	ASTM D1411 or BS 1377-3	As required by AGES-SP-01-006 Site Preparation, Earthworks and Bund Construction Specification Any change in material source or description and monthly during earthworks
Sulphate (SO_4) content	BS 1377-3	As required by AGES-SP-01-006 Site Preparation, Earthworks and Bund Construction Specification Any change in material source or description and monthly during earthworks
Maximum salt and organic matter content	ASTM C40/C40M	As required by AGES-SP-01-006 Site Preparation, Earthworks and Bund Construction Specification Any change in material source or description and monthly during earthworks
Field Density and moisture content: Sand Cone or Nuclear density (will be acceptable only after verification by parallel testing performed by Sand Cone, to the satisfaction of COMPANY.)	ASTM D1556 ASTM D6938	As required by Table 16.1 of AGES-SP-01-006 Site Preparation, Earthworks and Bund Construction Specification. Also see Note 1 below
CBR	BS 1377-4, BS 1377-9	Every 100 linear meters of each lane of final granular road layer only

NOTE 1: Field density tests shall be made for each lift of fill material placed and at the following minimum frequencies:

a. <u>Bulk Fill</u>

For mass fill, a minimum of one field density test shall be performed on every 1000 sq. m of each layer of compacted fill. For Structural and General fill, a minimum of one field density test for every 500 sq.m of each layer of compacted fill shall be performed.

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b. Bunds

For bunds, a minimum of one field density test per 100 meter length or 1000 sq.m, whichever is more, of each layer of compacted fill.

c. Tank Foundations

For tank foundations, a minimum of one field density test for every 200 sq. m with a minimum of five tests of each compacted layer.

d. Road Embankments

For road embankments, a minimum of one field density tests per 100 meter length or 1000 sq.m, whichever is more, of each layer of compacted fill.

e. Pipeline Bund

A minimum of one field density test for every 800 sq. m for the top layer and one field density test for every 1500 sq. m for each of other layers.

f. Pipeline ROW

A minimum of one field density test for every 500 linear m for the top layer and one field density test for every 1000 linear m for each of other layers.

The above soil test requirement is applicable during construction works. The subsurface soil test required prior/during construction works (Geotechnical) shall be in accordance with the requirements of AGES-SP-01-010, Onshore Geotechnical Investigation.

12 PAVING TESTING AND FREQUENCY

The CONTRACTOR shall be responsible for ensuring that the following paving tests are performed at the frequency shown below and in accordance with the requirements of AGES-SP-01-005 and the Standards listed in Section 2.0:

12.1 Bituminous Paving

MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
FINISHED PAVEMENT		
Density	ASTM D1188 and D2726	Test for every 1000m^2 of each layer
ASPHALT		
Sampling	AASHTO R 66	As required by individual test
Viscosity	ASTM D2170	Every day or 500 tons, whichever is more stringent

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MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
AGGREGATE		
Sampling	ASTM D75 and D3665	As required by individual test
Gradation	ASTM C117, C136 and D546	On each delivery to site/plant. One test per week of aggregate to be used.
Abrasion	ASTM C131	On each delivery to site/plant. One test per week of aggregate to be used.
Specific Gravity	ASTM C127, C128, C188 and D854	On each delivery to site/plant. One test per week of aggregate to be used.
Unit Weight	ASTM C29	On each delivery to site/plant. One test per week of aggregate to be used.
Soundness	ASTM C88	On each delivery to site/plant. One test per week of aggregate to be used.
HOT MIX		
Sampling	ASTM D979	As required by individual test
Marshall test	BS EN 12697-34	As required by individual test
Sieve Analysis	ASTM C136 and D546	Every 100m length of carriageway, or every 1000m^2 of area, whichever is more stringent.
Extraction	ASTM D2172	Every 100m length of carriageway, or every 1000m^2 of area, whichever is more stringent.
Density and stability	ASTM D2041	Every 100m length of carriageway, or every 1000m^2 of area, whichever is more stringent.
Voids	ASTM D3203	Every 100m length of carriageway, or every 1000m^2 of area, whichever is more stringent.
Recovery of Asphalt	ASTM D7906 & ASTM D5404 ASTM D1856	As Deemed necessary.

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MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
BITUMINOUS PAVING		
Stability and flow of samples taken before placing	Sampling: BS EN 12697-27 Testing: BS 594987, BS EN 12697-1, 2, 11, 14, 28 (compacted @ ±10°C of rolling temperature)	One sample every 100 tonnes or part thereof for each material. One sample shall be taken as the mean of three specimen plugs.
Stability and flow of samples cored from hardened course	Sampling: BS EN 12697-6; 10 Testing: BS 594987, BS EN 12697-1, 2, 11, 14, 28	One sample every 200m of road paving per lane, per layer or part thereof. One sample shall be taken as the mean of three specimen plugs.
Delivery and rolling temperatures	BS 598-1	Every truck delivery
AGGREGATES FOR ASPHALT		
Sieve analysis	BS EN 933-1	Weekly during asphalt production for each aggregate size in use. Sampling at asphalt plant.
Los Angeles abrasion of coarse aggregate	ASTM C131 and C535	Monthly during asphalt production for each coarse aggregate size in use. Sampling at asphalt plant.
BITUMEN		
Penetration	ASTM D5/D5M	One sample per week during asphalt production for each grade in use. Sampling at asphalt plant.

12.2 Interlocking Concrete Block Paving

MATERIAL AND REQUIREMENT	TEST METHOD	TEST FREQUENCY
Dimensions Compressive Strength Water Absorption	BS EN 1338	Every 5000m^2 of paved area

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13 ANCHOR BOLTS & NUTS

The CONTRACTOR shall be responsible for ensuring that the following tests are performed at the frequency shown below and in accordance with the requirements of AGES-SP-01-004 and the Standards listed in Section 2.0:

REQUIREMENT	TEST METHOD	TEST FREQUENCY
Chemical Composition	ASTM F1554	2% of Anchor from each vendor subject to a minimum of one (1) Number per diameter and type of bolt.
Mechanical Properties	ASTM F1554	2% of Anchor from each vendor subject to a minimum of one (1) Number per diameter and type of bolt.

14 POST INSTALLED ANCHORS

The CONTRACTOR shall be responsible for ensuring that the following tests are performed at the frequency shown below:

REQUIREMENT	TEST METHOD	TEST FREQUENCY
Material testing	Shall be as per Manufacturer's Test Certificates.	
Post Installation testing	Shall be as per as per Manufacturer's Specification.	

15 CONCRETE BATCH PLANT

Concrete batch plant shall be supplied and managed by the CONTRACTOR or a COMPANY approved Ready Mix Concrete Supplier.

15.1 Pre-construction inspection of concrete batch plant

The CONTRACTOR shall perform an initial inspection of the concrete production facilities and shall submit to the COMPANY a report that shall include as minimum the following:

a) Plant identification (owner, location and type).

b) Confirmation that batch plant is fully automated and fully computerized.

c) Nominal rated capacity.

d) Statement on whether or not the mixture trucks comply with the requirements of ASTM C94 and ACI 304 and note any exceptions.

e) Number, size, type and condition of delivery units.

f) Approved source and type of aggregate, cements, water, and admixtures.

g) Type and manufacturer of admixtures approved for use at the plant.

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h) Verification that all scales and dispensers have been calibrated.

i) Details of Adequate ice, chilled water, and/or liquid nitrogen facilities to meet hot weather production requirements.

15.2 In-process inspection of concrete batch plant

a. If CONTRACTOR is going to use his own batch plant, the CONTRACTOR shall provide experienced personnel to perform concrete batch plant inspection during the progress of the work. The duties of the batch plant inspection shall include, but not necessarily limited to the following:

b. The CONTRACTOR'S Batch Plant Manager will be responsible for the control of the quality of all batch plant operations.

c. The arrangements for inspection, testing, reporting and verification by QC department shall be approved by the COMPANY. As a minimum, the controls system implemented shall address the following:

i) Liaison with the site laboratory so as to ensure that the test status of materials is known.

ii) Supervision and recording of receipt, storage, holding and protection of materials.

iii) Supervision and recording of batching and mixing, including verification of batch computer print-outs before trucks are released.

iv) Control of moisture adjustments.

v) Initial and routine batch plant calibration and maintenance procedure.

vi) Regular reporting formats.

The QC department will verify that the control system is being implemented and verify the documentation raised, in accordance with an approved system.

d. If CONTRACTOR intends to use Ready Mix Concrete Supplier, then the CONTRACTOR is to seek COMPANY approval of SUPPLIER and ensure that all requirements of AGES-SP-01-001 Concrete Supply and Construction Specification are met.

15.3 Material Storage and Handling

a. Storage of cement shall be in accordance with the requirements of AGES-SP-01-001 Section 8.1.13.

b. Storage of aggregates shall be in accordance with the requirements of AGES-SP-01-001 Section 8.2.9.

15.4 Concrete Delivery

Concrete delivery shall be in accordance with the requirements of AGES-SP-01-001 Section 9.9.2.

16 ADDITIONAL SPECIFIC REQUIREMENTS

Not applicable.

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SECTION C – OTHER REQUIREMENTS

17 QUALITY PLANS

17.1 General

a. Quality plans shall address all aspects related to local conditions, such as climatic conditions, backup facilities, spare parts, transport possibilities, storage facilities, quarries, local MANUFACTURER/SUPPLIERs, test facilities (field laboratory).

b. Quality Management Systems shall comply with the applicable requirements of BS EN ISO 9001 "Quality Management Systems- Requirements" and BS EN ISO 9004, "Quality management – Quality of an organization – Guidance to achieve sustained success". Written quality plans and procedures shall be submitted to the COMPANY for review and approval.

c. Materials should be obtained from the same source(s) throughout the work (including both trial and production mixes).

17.2 Inspection, Testing and Reporting

a. Inspection and Test Plans (ITPs) identifying the tests, frequencies, acceptance criteria, and responsibilities shall be prepared as part of the quality plan.

b. A reporting system of Quality Control Records (QCRs) shall be part of the quality plan.

c. The reporting system shall record all results obtained during the testing.

d. Weather conditions shall be recorded in the quality records.

e. The proposed ITPs and QCRs shall be included in the Quality Plan.

f. Standard forms of inspection and test plans shall be used as a basis for the development of the ITPs and QCRs supplemented by the requirements of the Specifications for the work to be executed.

g. The COMPANY reserves the right to make inspections of the source of supply of materials.

h. Prior to supply of any material to site the CONTRACTOR shall obtain COMPANY approval by submitting necessary documentation such as test certificates, source of supply, date of manufacture, etc.

17.3 Quality Control and Assurance

a. CONTRACTOR shall be solely responsible for quality control of all materials and workmanship. The quality system shall provide for the planned and systematic control of all quality-related activities performed during design and construction of all components of the drainage system. Implementation of the quality system shall be in accordance with the Project Agreement, CONTRACTOR's approved Quality Manual and Project Specific Quality plan.

b. CONTRACTOR shall ensure that SUB-CONTRACTORS, SUPPLIERS and MANUFACTURERs have in place a QA/QC program which clearly establishes the authority and responsibilities of those responsible for the quality system. Persons performing quality functions shall have sufficient and well-defined authority to enforce quality requirements that they initiate or identify and to recommend

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and provide solutions for quality problems and thereafter verify the effectiveness of the corrective action.

c. Quality System and Quality Control requirements shall be identified and included in CONTRACTOR's Purchase documentation. Based on these requirements, the SUB-CONTRACTORS, SUPPLIERS and MANUFACTURERS will develop QA/QC programs which shall be submitted to COMPANY and CONTRACTOR for review and concurrence. The SUB-CONTRACTOR's, SUPPLIER's and MANUFACTURER's QA/QC programs shall extend to SUB-SUPPLIERS.

d. CONTRACTOR shall submit certified reports of all tests as soon as the tests are completed satisfactorily.

e. COMPANY reserves the right to inspect or to conduct a quality audit of materials, installation and workmanship standards and shall have unrestricted entry to the shop of the fabricator at all stages of manufacture and to witness any or all tests. SUB-CONTRACTORs, SUPPLIERs and MANUFACTURERs, within 30 days after award but prior to the pre-inspection meeting, shall provide CONTRACTOR and COMPANY with a copy of its Manufacturing and Inspection Plan for review and inclusion of any mandatory CONTRACTOR and COMPANY witness or hold points.

f. COMPANY may reject improper, inferior, defective, or unsuitable materials, installation and workmanship. All materials and workmanship rejected shall be repaired or replaced by CONTRACTOR as directed by the COMPANY.

18 SUB-CONTRACTORS, SUB-SUPPLERS

a. CONTRACTOR shall assume unit responsibility and overall guarantee.

b. CONTRACTOR shall transmit all relevant Purchase Order documents including COMPANY specifications to his SUB-CONTRACTORs and SUPPLIERs.

c. It is CONTRACTOR'S responsibility to enforce all Purchase Order and Project Specification requirements on its SUB-CONTRACTORS and SUPPLIERS.

d. CONTRACTOR shall submit all relevant SUB-CONTRACTOR and SUPPLIER engineering data to COMPANY.

e. CONTRACTOR shall obtain and transmit all SUB-CONTRACTOR and SUPPLIER warranties to COMPANY.

f. All drawings and engineering data submitted to COMPANY shall be in the English language, and in metric units.

19 NOTIFICATION OF NONCONFORMANCE

The CONTRACTOR shall notify the COMPANY within 24 hours of any test or any vendor supplied data that is required by this specification and does not meet the test limitations as shown in the following AGES:

Concrete Supply and Construction Specification	AGES-SP-01-001
Anchor Bolts and Grouting Specification	AGES-SP-01-004

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Roads and Paving – Design and Construction Specification	AGES-SP-01-005
Site Preparation, Earthworks and Bund Construction Specification	AGES-SP-01-006

20 DOCUMENTATION / MANUFACTURER DATA RECORDS

a. The CONTRACTOR and its SUBCONTRACTORs shall maintain test records and reports in accordance with the Standards as listed in Section 2.0.

b. The CONTRACTOR shall provide the COMPANY, within 24 hours, reports of all tests results.

c. The CONTRACTOR shall provide free access to COMPANY Representatives for supervision, inspection and qualitative check of the work and testing being executed.

d. In addition to test records and reports, the CONTRACTOR and its SUBCONTRACTORs shall review and maintain Manufacturer/Vendor data sheets, Material Certificates and Certificates of Conformance for the supply of cement and cement blends, reinforcing bars and fibers, admixtures, grouts and concrete coatings.

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SECTION D – STANDARD DRAWINGS & DATASHEETS

21 DATASHEET TEMPLATES

Not applicable.

22 STANDARD DRAWINGS

Not applicable.

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SECTION E - APPENDICES

Not Applicable

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